07 · L1 → L2 — Translation Challenges

> **Assignment:** *“Identify challenges in translation of the L1 to the L2: where did decisions have to be taken / gaps filled? What are the potential implications of these decisions?"*

> **Lecture Day 2:** *“Insufficient clarity of L1 ... Choices to be made when translating — aim of fidelity with underlying recommendations, but gaps to be filled / decisions to be taken. Need for clear documentation for validation + transparency."*

7.1 Why This Document Exists

No guideline is written in such a way that it can be translated into code without interpretation. Between L1 and L2 we had to take decisions at **nine points** that the source does not prescribe. Each of these decisions has clinical consequences — and is therefore a risk that ought to be disclosed.

We classify the gaps by type:

Type	Meaning
Context gap	The source presupposes conditions that are not given in our setting
Missing threshold	The source names no criterion, so we had to set one
Implicit knowledge	The source presupposes clinical judgement that software does not have
Vagueness	The source is phrased indeterminately
Evidence gap	The available evidence supports our application only to a limited extent

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7.2 The Nine Challenges

C-01 · A Professional Instrument Becomes a Lay Instrument

Type	Context gap
Gap in L1	Both the CDC criteria and the Southampton grading were developed for assessment by **trained personnel**. Our users are medical laypersons assessing their own wound
Our decision	Translation of the visual grading into an **image selection** (“Which picture fits best?") instead of descriptive questions
Rationale	Images overcome language and educational barriers. A patient can answer “my wound looks like picture 3" more reliably than “is there erythema with signs of inflammation?"
Implication	Loss of granularity compared with the original. The assignment depends on the quality and representativeness of the graphics — particularly for different skin types, in which erythema appears visually very different
Risk	🔴 **high** — the largest untested risk of the entire approach
Affected	DE-030, DE-035, R3, R4, R6a, R6b
Mitigation	Worst-first logic; the wound appearance “open/pus" **always** escalates; residual risk explicitly documented (10 QA)

C-02 · Southampton Sub-Grades Without Centimetre Measurements

Type	Implicit knowledge
Gap in L1	Southampton distinguishes "≤ 2 cm" from "> 2 cm along the wound" (sub-grades IIIa/b, IVa/b). Patients do not measure their wound at home
Our decision	Replaced by the qualitative question *"At a small spot or along the whole wound?"* (DE-037)
Rationale	A semantically close approximation without a measuring instrument. The relative assessment "localised vs. extensive" can also be made reliably by laypersons
Implication	Lower precision than the original. Possible systematic **underestimation**, because "small" is interpreted generously in subjective terms
Risk	🟡 **medium**
Affected	DE-037, R11

C-03 · From What Point Is Erythema No Longer "Normal Post-Operatively"?

Type	Missing threshold
Gap in L1	Neither the CDC nor NICE nor Southampton define from which post-operative day erythema is considered pathological. Southampton has no time dimension at all
Our decision	**Day 3** as the boundary: mild signs of inflammation (Southampton I) are GREEN up to and including day 3, and YELLOW from day 4 onwards (R6a). **Pronounced** signs (grade II) are YELLOW from day 1 onwards (R6b)
Rationale	The physiological inflammatory phase of wound healing lasts about 2–3 days. The value is based on wound healing physiology, but was set by us
Implication	Chosen too early → false alarms during the normal inflammatory phase, loss of trust, alert fatigue in the practice. Chosen too late → delayed detection of early infections
Risk	🔴 **high**
Affected	`normal_healing_days`, R6a, R6b
Note	The split into R6a/R6b only came about through Bug B-01 — originally the entire rule was gated, with the result that even **pronounced** signs of inflammation on days 1–2 came out GREEN

C-04 · Scab — Warning Sign or Reassurance?

Type	Vagueness
Gap in L1	A scab/crust is not listed as a warning sign in any of our sources — but neither is it explicitly named as normal. Patients regularly interpret it as a problem
Our decision	A dedicated data element (DE-041) with a **reassuring informational rule** I4. Not a warning sign
Rationale	Avoids unnecessary contacts and addresses a real concern documented in Persona 1. A CDSS that only warns gives away half of its value
Implication	Risk of false reassurance if the scab is part of an infectious process. Safeguarded by worst-first: warning rules always override the notice
Risk	🟢 **low**
Affected	DE-041, I4
Verified in	T9, T10

C-05 · CDC Criteria Presuppose a Medical Examination

Type	Context gap
Gap in L1	The CDC/NHSN definition rests in part on criteria that cannot be collected at home: culture results, active reopening of the wound by the physician, imaging procedures, diagnosis by a physician
Our decision	Adoption of **the clinical-visual criteria only**: purulent drainage, localized pain/tenderness, swelling, erythema, heat
Rationale	Only these can be captured through self-observation
Implication	**WundCheck cannot diagnose an SSI** — it detects a suspicion and triages. All outputs are accordingly phrased as recommendations for action ("contact your…"), not as a diagnosis. Cases that manifest primarily through laboratory values or imaging are systematically not captured
Risk	🟡 **medium**
Affected	the entire output logic, all output wordings

C-06 · The Evidence Base Supports Our Instrument Only to a Limited Extent

Type	Evidence gap
Gap in L1	Campwala et al. (2019) found only **limited predictive value** for Southampton (n = 22, retrospective, implant-based breast reconstruction) — a population that deviates markedly from our scope. ASEPSIS performed better
Our decision	Southampton is used as a **communication and triage grid**, not as a prognostic instrument. ASEPSIS was **rejected** despite its better figures
Rationale	ASEPSIS requires antibiotic administration, drainage, debridement, hospital stay and culture results over several days — entirely uncollectable for patients at home. We choose the applicable option over the statistically better one
Implication	We can make **no statement about the course of healing**, only about the current need for action. The transferability of the study results to our population is limited in any case
Risk	🟡 **medium**
Affected	the entire L1 selection, communication of the value proposition

C-07 · Fever Thresholds for Two Urgency Levels

Type	Missing threshold
Gap in L1	For deep incisional SSI the CDC names "fever (>38 °C)" — but **one** threshold. We need **two**, because we distinguish between "contact the practice today" (orange) and "emergency immediately" (red)
Our decision	`fever_orange = 38.0` (adopted from the CDC) · `fever_red = 39.0` (set by us). In addition: chills trigger RED independently of the temperature
Rationale	39 °C as the emergency threshold follows common clinical convention. Chills were added because they can precede measurable fever and are a sepsis warning sign
Implication	The 39 °C limit is **not source-based**. In older or immunosuppressed patients a severe infection can run its course without a high fever — these cases are partly caught by R9 (feeling of illness)
Risk	🟡 **medium**
Affected	`fever_red`, R1, R5

C-08 · No Trajectory on the First Day

Type	Context gap

Gap in L1	The CDC phrases criteria as *"new or **worsening** localized pain or tenderness"* — deterioration presupposes a prior finding. On first use there is no previous day
Our decision	Trajectory-dependent rules (the R5 sub-condition, R12) only become active from the second check day. On the first day `pain_trend` is collected, but interpreted as a self-assessment relative to the day of surgery
Rationale	Without a baseline, a statement about a trend is not possible. The alternative — dispensing with trajectory rules — would have rendered a substantial part of the CDC criteria unusable
Implication	On the first day of use, sensitivity is lower. Patients who only start using WundCheck late additionally lose trajectory information. This increases the importance of adherence to use (`check_streak_days` as a monitoring indicator)
Risk	🟢 low
Affected	DE-024, DE-047, DE-048, R5, R12

C-09 · Severity scale ≠ urgency of action

Type	Vagueness
Gap in L1	The Southampton system grades the **severity of the finding** (grades IV and V are "major complications"). It says nothing about **how quickly** anyone has to act. Our traffic light, however, encodes exactly that: urgency
Our decision	Severity and urgency are decoupled. Isolated purulent discharge (grade IV) is ORANGE — "contact your practice today" — not RED. RED is reserved for systemic signs (R1), lymphangitis (R2) and dehiscence or the "open / pus" wound image (R3)
Rationale	Purulent discharge is the primary criterion of a superficial incisional SSI and does require treatment — but usually not the emergency service at night. A RED classification that sends every superficial SSI to A&E would overload emergency care and devalue the RED level for the cases that genuinely need it
Implication	Our traffic light is **not** a Southampton grade. Anyone reading "grade IV" in the FHIR export must not infer "emergency". Both values are therefore exported separately
Risk	🟡 medium
Found by	Test case T15 — see bug B-06

7.3 Overview Table

ID	Short description	Type	Risk	Affected rules
C-01	Professional instrument → lay instrument	Context gap	🔴 high	R3, R4, R6a, R6b
C-02	Sub-grades without centimetres	Implicit knowledge	🟡 medium	R11
C-03	Boundary of "normal" erythema = day 3	Missing threshold	🔴 high	R6a, R6b
C-04	Scab as a notice, not as a warning	Vagueness	🟢 low	I4
C-05	Only visual CDC criteria usable	Context gap	🟡 medium	all outputs
C-06	Southampton as triage, not prognosis	Evidence gap	🟡 medium	L1 selection
C-07	Two fever thresholds instead of one	Missing threshold	🟡 medium	R1, R5
C-08	No trajectory on the first day	Context gap	🟢 low	R5, R12
C-09	Severity scale ≠ urgency of action	Vagueness	🟡 medium	R3, R4

7.4 What Follows From This

Two of the three high-risk decisions concern the same point: the translation of a clinical finding into a lay assessment. That is precisely where our tests also found the most serious errors (B-01, B-03). This is no coincidence — it is the structural weak point of every patient-facing CDSS.

All eight decisions would require clinical sign-off. Lecture Day 5 names *"clinical expert sign-off on rule logic and edge cases"* as a characteristic of adequate validation. In our prototype it is absent — the decisions were taken by a group of students. We list this as a **limitation**, not as a completed item.

The documented thresholds are the lever for a review. Because all threshold values reside as named constants in the L3 (`meta.thresholds`), a clinical professional can review and change them without understanding the code. The documentation of these challenges is therefore not merely a submission requirement, but the concrete working basis for precisely this review.

Related documents: 03 L1 · 06 Decision Logic · 10 QA